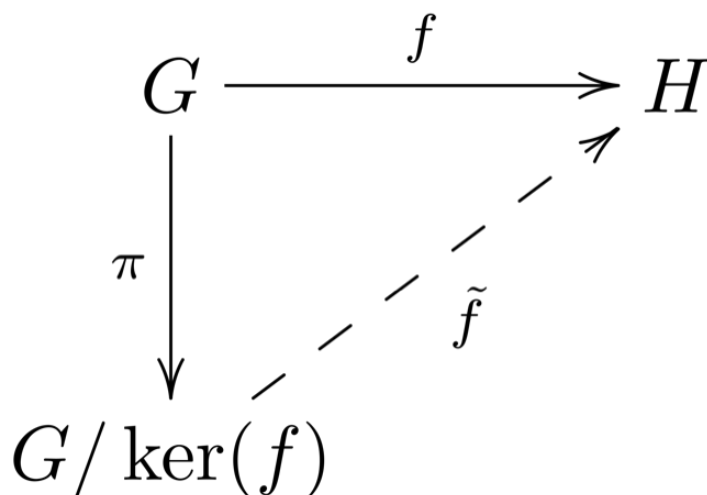


Chapter 5. Factor groups

5.1 Normal subgroups



In this video, we will talk about a particular class of normal subgroups.

Kernels

PROPOSITION 5.4. Let $\varphi : G \rightarrow G'$ be a group homomorphism. Then the kernel of φ , defined as,

$$\ker \varphi = \{k \in G \mid \varphi(k) = 1_{G'}\}$$

is a normal subgroup of G . Moreover, φ is a monomorphism if and only if $\ker \varphi = \{1_G\}$.

Proof

We first prove that the kernel of φ is a subgroup of G . Using the subgroup Test, we need to check the following:

•) $1_G \stackrel{?}{\in} \ker \varphi$

$$\varphi(1_G) = 1_{G'} \quad \checkmark$$

•) $k, k' \in \ker \varphi \stackrel{?}{\Rightarrow} kk' \in \ker \varphi$

$$\varphi(kk') = \varphi(k)\varphi(k') = 1_{G'} 1_{G'} = 1_{G'} \quad \checkmark$$

$$\bullet) K \in \text{Ker } \varphi \stackrel{?}{\Rightarrow} K^{-1} \in \text{Ker } \varphi$$

$$\varphi(K^{-1}) = \varphi(K)^{-1} = 1_{G'}^{-1} = 1_{G'}$$

By the Subgroup Test, we have that $\text{Ker } \varphi$ is a subgroup of G . To check that $\text{Ker } \varphi$ is normal, we need to prove that $g \text{Ker } \varphi g^{-1} \subseteq \text{Ker } \varphi$ for all $g \in G$.

Take $g \in G$ and $K \in \text{Ker } \varphi$, and let us prove that $gKg^{-1} \stackrel{?}{\in} \text{Ker } \varphi$.

$$\varphi(gKg^{-1}) \stackrel{?}{=} 1_{G'}$$

$$\begin{aligned}\varphi(gkg^{-1}) &= \varphi(g)\varphi(k)\varphi(g^{-1}) \\ &= \varphi(g)1_{G'}\varphi(g)^{-1} = 1_{G'}\end{aligned}$$

$$\Rightarrow gkg^{-1} \in \text{Ker } \varphi$$

By the Normality Test we have that
 $\text{Ker } \varphi \triangleleft G$.

It remains to show that

$$\varphi \text{ is a monomorphism} \iff \text{Ker } \varphi = \{1_G\}.$$

$\Rightarrow$ | Suppose that φ is a monomorphism
and take $k \in \text{Ker } \varphi$.

$$\underline{k \in \text{Ker } \varphi} \Rightarrow \varphi(k) = 1_{G'}$$

We also have that $\varphi(1_G) = 1_{G'}$.

So, $\varphi(k) = 1_{G'} = \varphi(1_G)$ and

since φ is injective we obtain that

$k = 1_G$. This shows that

$$\text{Ker } \varphi = \{1_G\}.$$

$\Leftarrow$ | Suppose now that $\text{Ker } \varphi = \{1_G\}$ and let us show that φ is injective.

If $g_1, g_2 \in G$ are such that $\varphi(g_1) = \varphi(g_2)$. Then we need to show that $g_1 = g_2$.

$$\varphi(g_1) = \varphi(g_2) \Rightarrow \varphi(g_1) \varphi(g_2)^{-1} = 1_{G'}$$

$$\Rightarrow \varphi(g_1) \varphi(g_2^{-1}) = 1_{G'} \Rightarrow$$

$$\Rightarrow \varphi(g_1 g_2^{-1}) = 1_{G'} \Rightarrow g_1 g_2^{-1} \in \text{Ker } \varphi$$

$$\Rightarrow g_1 g_2^{-1} = 1_G \Rightarrow \underline{g_1 = g_2}$$

